MIL-S-14617C(AL)
16 May 1985
SUPERSEDING
MIL-S-14617B (WC)
12 April 1972

# MILITARY SPECIFICATION

## SPRAY GUN, PAINT, TOUCH-UP

This specification is approved for use by the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of air-operated, touch-up paint spray gun.

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-12651 - Sprayers, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Service.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Temperature

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete spray gun for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design and construction. The spray gun shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The gun shall be designed for hand operation and shall be suitable for light, fine painting, decorating, and stenciling. The spray gun shall be a bleeder or non-bleeder, pressure or siphon feed type with an internal or external mix cap and a siphon cup. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the gun or any gun component. The gun shall be furnished complete so that it can be used for the operations specified herein.

* 3.2.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are



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used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889. Parts subject to contact with the sprayable materials specified in 3.3.1 shall be inert to chemical reaction with those materials.

* 3.2.1.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

* 3.2.2 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.2.4 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

* 3.2.5 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

* 3.2.6 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.2.7 Protective finish. Exposed metal (see 6.4.1) and working surfaces, except when the parts involved are made of corrosion resisting materials, shall be coated or covered with a painted aluminum finish or with an electrochemical finish.

3.3 Performance and product characteristics.

3.3.1 Sprayable materials. The spray gun shall spray enamels, lacquers, primers, shellacs, varnishes, stains, oil paints, and related fluids with a viscosity range of 10 seconds to 25 seconds using a No. 4 Ford cup at 77° Fahrenheit (F) ±2°F. The gun shall satisfactorily spray these materials on vertical surfaces without spitting, spattering, fogging, or leaking.



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* 3.3.2 Spray patterns. The spray gun shall produce a round pattern with the pattern control valve closed and a symmetrical fan spray pattern with the pattern control valve open (see Figure 1, normal spray patterns). The spray pattern shall rotate but not change in shape when the air cap is rotated. The spray gun, using sprayable materials in accordance with 3.3.1 and held stationary three inches from, and perpendicular to, the surface to be sprayed, shall produce spray patterns as follows:

(a) A round pattern having a maximum effective diameter of 1/2 inch.
(b) A fan shaped pattern having minimum effective dimensions of 1 3/4 inches along the major axis and one inch along the minor axis.

The spray gun shall not produce any of the defective spray patterns shown in Figure 1 i.e., a heavy top or bottom pattern, a heavy right or left side pattern, a heavy center pattern, or a split pattern.

3.3.3 Air consumption. The spray gun shall produce the normal spray patterns specified in 3.3.2 with air consumption of 1.5 to 4.5 cubic feet per minute at 30 to 60 pounds per square inch gage (psig).

3.3.4 Operational endurance. The spray gun and all components shall withstand without damage and operate satisfactorily after 10,000 actuations (full stroke) of the trigger while under the pressures specified in 3.3.3.

3.3.5 Tightness.

* 3.3.5.1 Overall leakage. The spray gun, with the outlet ports blocked and the trigger in the spraying position, shall withstand an air pressure of 80 $\pm$ 2 psig for 60 seconds with not more than a 5 psig pressure drop.

3.3.5.2 Air valve leakage. The spray gun shall withstand an air pressure of 80 $\pm$2 psig at the air valve for one minute with not more than a 5 psig pressure drop due to leakage.

3.4 Components.

3.4.1 Body. The spray gun body shall be of drop-forged aluminum. The spray head shall support and provide correct alignment of the nozzle assembly.

3.4.2 Nozzle assembly. The nozzle assembly shall consist of an air cap, fluid nozzle, and required auxiliary parts. The nozzle assembly shall be suitable for producing the normal spray patterns specified in 3.3.2, using the spray material and the air consumption specified in 3.3.1 and 3.3.3. The nozzle shall be adjustable from a horizontal to a vertical spray pattern.

3.4.2.1 Air cap. The air cap shall be secured to the spray gun head by a retainer ring, or other suitable means, and shall be self centering. The air cap shall be capable of being removed and replaced to permit cleaning and repair. The cap shall be fabricated from either a brass or steel forging or rod, or from an aluminum or zinc die casting.



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3.4.2.2 Fluid nozzle. The fluid nozzle shall incorporate the required orifice to produce the normal spray patterns specified in 3.3.2, using the spray material and air consumption specified in 3.3.1 and 3.3.3. The fluid nozzle shall be made of carbide or stainless steel.

3.4.2.3 Auxiliary parts. Baffle plates, tension washers, retainer rings, and other auxiliary parts shall be capable of meeting the requirements specified herein.

3.4.3 Valves. The spray gun shall have a fluid valve, an air valve, and a pattern control valve. The valves shall be removeable and replaceable to facilitate cleaning and repair. The fluid valve and the air valve shall be held normally closed by coil springs enclosed within the spray gun body to provide shut-off of fluid and air flow through the spray gun. Valve packings shall be graphited fiber, treated leather, or synthetic material.

3.4.3.1 Fluid valve. The fluid valve shall be a needle type made of stainless steel. The valve shall be in contact with an adjusting screw mounted at the rear of the spray gun body to permit changing the quantity of fluid flow through the fluid orifice when the fluid valve is opened.

3.4.3.2 Air valve. The air valve shall incorporate a stem, made of stainless steel, to open the air valve when the trigger is actuated.

3.4.3.3 Pattern control valve. The pattern control valve shall permit changing the spray pattern from round to fan-shaped and from fan-shaped to round (see 3.3.2).

3.4.4 Trigger. The trigger shall permit operation by the operator's forefinger to provide extremely sensitive control of the spray. It shall control both the fluid valve and the air valve and shall operate to permit air flow before fluid flow and fluid shut off before air shut off.

3.4.5 Connections. The spray gun shall be provided with a fluid connection and an air connection. The fluid connection shall be located below the spray head and shall be suitable for mounting the siphon cup specified in 3.4.6. The air connection shall be a detachable 1/4-18 National Pipe Thread (NPT) with a 60° bevel taper seat. The air connection shall be fabricated from drawn brass rod, or other suitable material, and shall be located on the spray gun body so it shall not interfere with operation of the spray gun.

3.4.6 Siphon cup. The spray gun shall be equipped with a siphon cup having a capacity of not less than four ounces, mounted below the spray head. The siphon cup cover shall be of metal not less than 0.035 inch thick, have a vent not less than 0.0625 inch in diameter, and be detachable from the spray gun. The cup shall be of glass or aluminum and shall be detachable from the cup cover. The outside diameter of the cup shall be not less than two inches. The throat diameter shall be not less than 1.5 inches. The seals between the spray gun and the cup cover and between the cup cover and the cup shall be fluid tight.

3.5 Product identification. The spray gun shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.



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* 3.6 Workmanship. Standards of workmanship shall assure that the spray gun shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (d) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Leakage tests (see 4.6.2.1 and 4.6.2.2).
(c) Spray pattern test (see 4.6.3).
(d) Packaging inspection (see 4.7).

* 4.4.1 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.

* 4.5 Product examination. The spray guns shall be examined to determine compliance with the requirements of Section 3 and the contract. As a minimum, the following characteristics shall be examined:

| CHARACTERISTICS | REQUIREMENT PARAGRAPH |
|---|---|
| Design and construction | 3.2 |
| Materials | 3.2.1 |



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| | |
|---|---|
| Chemically inert parts | 3.2.1 |
| Reclaimed materials when applicable | 3.2.1.1 |
| Interchangeability | 3.2.2 |
| Threads | 3.2.3 |
| Castings and forgings | 3.2.4 |
| Welding, brazing, and soldering | 3.2.5 |
| Fastening devices | 3.2.6 |
| Protective finishes | 3.2.7 |
| Performance and product characteristics | 3.3 |
| Sprayability | 3.3.1 |
| Spray patterns | 3.3.2 |
| Air consumption | 3.3.3 |
| Operational endurance | 3.3.4 |
| Overall leakage | 3.3.5.1 |
| Air valve leakage | 3.3.5.2 |
| Components | 3.4 |
| Body | 3.4.1 |
| Nozzle assembly | 3.4.2 |
| Air cap | 3.4.2.1 |
| Fluid nozzle | 3.4.2.2 |
| Auxiliary parts | 3.4.2.3 |
| Valves | 3.4.3 |
| Fluid valve | 3.4.3.1 |
| Air valve | 3.4.3.2 |
| Pattern control valve | 3.4.3.3 |
| Trigger | 3.4.4 |
| Connections | 3.4.5 |
| Siphon cup | 3.4.6 |
| Product identification | 3.5 |
| Workmanship | 3.6 |

A certificate of conformance may be used to satisfy examination requirements when more than a visual or dimensional verification would be required.

4.6 Tests.

4.6.1 Operational endurance test. The spray gun shall be subjected to 10,000 operation cycles of the trigger while under an air pressure of 30 to 60 psig. On conclusion of the test, the spray gun shall be inspected for damage and tested in accordance with 4.6.2 and 4.6.3.

4.6.2 Leakage tests.

4.6.2.1 Air valve test. The air valve shall be tested for leakage by connecting an air pressure gage (with an accuracy within 0.25 percent) to the spray gun air inlet connection and attaching a 20 foot length of 0.3125 inch, inside diameter, air hose between the pressure gage and the air pressure source. An air pressure of 80 ±2 psig shall be applied and entrapped between the spray gun and the air source shut-off valve for a period of one minute. The pressure gage shall be observed to determine compliance with 3.3.5.2.

* 4.6.2.2 Over-all leakage test. The air ports in the air cap shall be blocked by fitting the spray gun with a blank air cap, or by other suitable means, employing the standard air cap retainer or connection. Cap or plug the fluid



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inlet passage. An air pressure gage (with an accuracy within 0.25 percent) shall be connected to the spray gun air inlet connection and a 20 foot length of 0.3125 inch, inside diameter, air hose shall be attached between the pressure gage and the air pressure source. With the spray gun trigger held in the spraying position (opened) and the spray gun pattern control valve in the open position, a pressure of 80 ±2 psig shall be applied and entrapped between the spray gun and the air source shutoff valve for a period of 60 seconds. The pressure gage shall be observed to determine compliance with 3.3.5.1.

4.6.3 Spray pattern test. The spray gun siphon cup shall be filled with fluid having a viscosity of 10 seconds, using a No. 4 Ford cup at 77° ± 2°F. The spray gun shall be connected to an air source providing the air consumption requirements specified in 3.3.3. The spray gun shall be held perpendicular to, and at a distance of not more than three inches from, the surface to be sprayed. The patterns shall be sprayed with the pattern control valve opened and closed to determine compliance with 3.3.2. The spray gun shall be checked for producing horizontal and vertical patterns by rotating the air cap in 90° increments through 360°. This test shall be repeated using a fluid with a viscosity of 25 seconds and a No. 4 Ford cup at 77° ± 2°F.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-S-12651.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The paint spray gun covered by this specification is intended for intermittent duty in delicate and intricate spray work involving fine finishes, blending, stenciling, spotting, feathering repaired areas on small items, and decorative work.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) First article when required (see 3.1).
(c) Product identification if different (see 3.5).
(d) First article inspection if different (see 4.3).
(e) Quality conformance inspection if different (see 4.4).
(f) Sampling when required (see 4.4.1).
(g) Level of preservation, packing, and marking if different (see 5.1 and 5.2).



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6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

* 6.4 Definitions.

6.4.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, compressed air, solvents, or the material (paint) being sprayed (see 3.2.7).

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

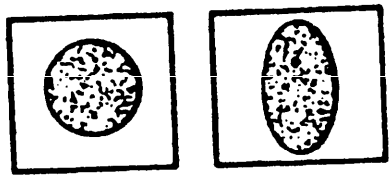
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NORMAL SPRAY PATTERNS



DEFECTIVE SPRAY PATTERNS

FIGURE 1. Spray patterns.



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| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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